Table I: Decision scope of representative approaches and the present framework. The rows compare mechanisms, not performance under a common benchmark.

Approach	Principal mechanism	Connection to the present decision problem
CPPS malware analysis [3], [5], [6]	Spreading, incubation, coupling and protection priorities	Establishes the cross-layer consequence and fixed-ranking comparators.
Moving-target defense [28], [29], [37]	Changes system or control characteristics to impede attacks	Demonstrates adaptive defense with a different intervention space.
Dynamic-botnet resiliency [30]	Coordinates cyber and physical resilience under botnet attacks	Shares the cross-layer motivation; attack models and actions differ.
DRAD-CF-MPC	Same-state previews of interpretable node sets with a CDP-referenced gate	Links infection-aware candidates, predicted consequence and acceptance evidence; matched controls separate their effects.

Table II: Comprehensive candidate policy families. The number of executable candidates is state/variant dependent (8–12 in the separate rolling records).

Policy	Main criterion	Engineering interpretation
NP	No protection	Unprotected reference baseline.
RP	Random selection	Non-informative defense without cyber or physical knowledge.
CDP	Communication degree	Protects communication hubs to suppress malware diffusion.
PDP	Power-layer importance	Protects nodes associated with physically important buses.
CP	Capacity-proxy protection	Protects nodes coupled to high-capacity or high-impact buses.
PHYSICS	Physical-impact score	Protects nodes whose coupled buses have high physics-informed impact scores.
HYBRID	Cyber–physical weighted score	Balances communication centrality and physical consequence.
DRAD-score	Dynamic risk-aware score	Uses topology, physical consequence, infection pressure, and information exposure.
Rank-ensemble	Rank aggregation	Combines cyber, physical, and infection-based rankings.
CDP-PHYSICS / CDP-PDP	CDP plus physical/PDP supplement	Keeps communication-hub coverage while adding physical-consequence awareness.

Table III: Reference configuration for the comprehensive policy-library assessment. Rolling and local-information protocols specify their own rollout counts and risk profiles.

Category	Setting	Description
Physical network	IEEE 118-bus system	Benchmark power system for physical consequence and restoration evaluation.
Communication network	118-node BA scale-free network	Heterogeneous cyber topology for malware diffusion.
Coupling patterns	DCAC, DCDC, DDAC, DDDC, RC	Degree-correlated, degree-disassortative, and random mappings.
Defense strategies	NP, RP, CDP, PDP, CP, DRAD-score, DRAD-CF-MPC	Reference policies and the proposed Safe-Guarded controller.
Attacker information	0.25, 0.50, 1.00	Limited, partial, and full structural knowledge.
Defense budget	0, 20, 40, 60, 80, 100	Number of protected cyber nodes.
Attack onset time	0, 30, . . ., 300	Physical activation time after malware propagation.
Monte Carlo repetitions	50 per task	Repeated stochastic malware simulations in the comprehensive-library results.
Counterfactual rollouts	$N_{\text{cf}} = 20$	Common-random-number scenario previews used for candidate comparison.
Rollout horizon	Onset-aligned malware simulation horizon	Candidate policies are evaluated from the observed pre-attack state to the physical activation time.
State-risk coefficients	$\theta_I = 1, \theta_E = 0.75, \theta_N = 0.5$	Fixed SEID severity factors used in Eq. (17).
DRAD-score weights	ρ_1=0.20, ρ_2=0.15, ρ_3=0.15, ρ_4=0.15, ρ_5=0.20, ρ_6=0.15	Fixed weights in Eq. (15); communication degree and infection pressure prioritized, physical topology features secondary.
Objective weights	$\lambda_{\text{EP}} = 0.68$, $\lambda_{\text{PAS}} = 0.24$, $\lambda_{\text{RT}} = 0.06$, $\lambda_{\text{RI}} = 0.02$	Fixed weights used for same-state counterfactual comparison.
Safe-Guard parameters	$\gamma = 0.10$, $\kappa = 0.50$	Switching margin and conservativeness coefficient.
Numerical stabilizer	$\varepsilon_0 > 0$	Machine-level constant used only to avoid zero denominators.
Bus-impact weights	$\omega_L = 0.35$, $\omega_C = 0.25$, $\omega_k = 0.20$, $\omega_b = 0.20$	Fixed weights in Eq. (8); load-demand prioritized, topology features secondary.
RT normalization	$T_{\text{rec}}^{\max} = 21$	Restoration-time normalization in Eq. (21).
Primary metrics	$\text{EP}_{\max}$, PAS, RT, RI, RE	Degradation and restoration indicators.

Table IV: Comprehensive policy-library scenario-preview comparison. EP is the success-discounted composite score; PAS and RT/RI/RE are service and algebraic recovery proxies.

Strategy	Mean EP_{max} ↓	Mean PAS↓	Mean RT↓	Mean RI↑	Mean RE↑	EP gain vs NP	EP gain vs CDP
DRAD-CF-MPC	**0.289**	**0.306**	**6.638**	**0.827**	**0.444**	**63.2%**	**3.0%**
CDP	<u>0.298</u>	<u>0.315</u>	<u>6.765</u>	<u>0.822</u>	<u>0.439</u>	<u>62.1%</u>	–
DRAD-score	0.348	0.369	7.657	0.796	0.402	55.7%	−16.8%
CP	0.465	0.499	9.697	0.716	0.319	40.8%	−56.0%
PDP	0.465	0.503	9.770	0.714	0.316	40.8%	−56.0%
RP	0.567	0.611	11.529	0.647	0.244	27.9%	−90.3%
NP	0.786	0.848	17.968	0.449	0.152	–	−163.8%

Note: Arrows indicate the preferred direction. Bold and underlined values denote numerical best and second-best results, respectively, within this scenario-preview comparison; they do not indicate statistical significance. Failed attacks already enter PAS as zero, so EP discounts success frequency twice.

Table V: Scenario-level gains over CDP for the comprehensive policy library. The 6.20% mean of relative scenario gains differs from the 3.0% gain of pooled means.

Coupling pattern	Mean EP gain	Median EP gain	Min. EP gain	Max. EP gain	Positive cases
DCAC	5.06%	4.60%	−9.13%	19.37%	72.22%
DCDC	5.28%	2.69%	−8.58%	27.04%	55.56%
DDAC	**7.97%**	**9.69%**	−10.36%	23.38%	72.22%
DDDC	5.41%	1.38%	−22.46%	**34.34%**	50.00%
RC	7.29%	3.00%	−25.28%	29.46%	72.22%
Overall	**6.20%**	**3.77%**	−25.28%	**34.34%**	**64.44%**

Note: Positive cases are scenario-budget combinations in which the composite EP score is lower than CDP. A positive fraction is descriptive and is not a significance test. Overall values summarize a different aggregation than pooled strategy means.

Table VI: Budget-dependent comparison for the comprehensive policy library. Bold metric entries identify the lower value within the same-budget CDP/adaptive pair; bold EP gain marks the largest gain across budgets.

Defense budget	CDP EP↓	DRAD-CF-MPC EP↓	EP gain	CDP PAS↓	DRAD-CF-MPC PAS↓	CDP RT↓	DRAD-CF-MPC RT↓
0	0.786	0.786	0.0%	0.848	0.848	17.968	17.968
20	0.466	**0.451**	3.3%	0.496	**0.483**	10.217	**9.981**
40	**0.183**	0.185	−1.0%	**0.187**	0.190	**4.213**	4.265
60	0.132	**0.122**	7.7%	0.132	**0.123**	3.093	**2.938**
80	0.125	**0.114**	8.7%	0.125	**0.115**	2.799	**2.652**
100	0.096	**0.077**	**20.0%**	0.100	**0.079**	2.298	**2.025**

Note: Ties are unbolded. Boldface is numerical and does not indicate significance. A negative EP gain means CDP outperforms DRAD-CF-MPC under that budget. EP gains are computed from unrounded simulation outputs; the tabulated EP, PAS, and RT values are rounded to three decimals.

Table VII: Baseline component-removal comparison for the comprehensive policy library. The matched rolling validation below provides a complementary controlled attribution analysis.

Variant	Mean EP_{max} ↓	Gain vs. NP
NP	0.786	0.0%
RP	0.567	27.9%
CDP	0.298	62.1%
DRAD-score	0.348	55.7%
w/o MPC	0.341	56.6%
w/o physical consequence	0.318	59.5%
w/o Safe-Guard	0.307	61.0%
Full DRAD-CF-MPC	**0.289**	**63.2%**

Table VIII: Additional matched rolling comparisons for the comprehensive policy library, using a distinct continuation protocol from Table VII. Gain is PAS of reference B minus PAS of method A, so positive gain favors A.

Method A	Reference B	Mean PAS A	Mean PAS B	Gain [95% interval]	Holm p
FULL	CDP	.483687	.482881	$-.000806\ [-.001815, .000198]$	.509
FULL	No dynamic risk	.483687	.483691	$.000003\ [-.000421, .000610]$	1.000
FULL	No guard	.483687	.464499	$-.019188\ [-.025131, -.014649]$	1.72×10^{-16}

400 paired setting–seed cases, fixed eight settings, three decision stages. "No guard" removes the acceptance gate but retains rollout selection. The complete six-row comparison is provided in Supplementary Material Table S5. Intervals and tests follow Section V-C; positive-gain and low-p claims must be read in the sign convention above.

Table IX: Separate PGLib electrical checks, retaining all 18 cases per system. Screened branches pass a conservative postcheck filter; the last column does not count all ratings provided by the source.

System	Checked success	Solver failure	Island	Success, partial postcheck	Screened / active branches
IEEE14	17	1	0	0	20/20
IEEE57	15	3	0	0	80/80
IEEE118	18	0	0	0	186/186
IEEE300	0	7	5	6	403/411